PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis
AUTHORS	Hirt, Julian; Langer, Gero; Wilde, Fabian; Bauernschmidt, Dorothee; Meyer, Gabriele; Bieber, Anja

VERSION 1 – REVIEW

REVIEWER	Struckmeyer, Kristopher Oklahoma State University, Human Development and Family Science
REVIEW RETURNED	30-Jul-2021

GENERAL COMMENTS	This was a very interesting protocol and I am very intrigued to see the results from your QCA and meta-analysis. Some minor corrections: Page 9, Line 3 "Gender differences have..." - I would remove this sentence or go more in-depth regarding the known gender differences. In its current state, it seems out of place and does not seem to provide much information. Page 9, Line 9 "The worldwide COVID-19..." - change "e.g." to "related to". I noticed throughout the paper that you use "e.g." frequently. I would suggest placing those statements in parentheses "(e.g., lack of information...) or changing the wording to "related to" or something to similar to help with sentence flow. Page 9, Line 33 "...to cope with emotional issue" - insert "an" between "with" and "emotional issue" Page 10, Line 7 "...telephone counseling includes..." - This sentence seems out of place. I would either remove or move to another part of the paper. Page 10, Line 18 "...for a more timely and individual..." - change individual to individualized Page 10, Line 35-39 "The authors concluded..." - delete ", which" and change influence to influenced. Page 10, Line 43 "completing strategies for services" - please clarify this statement. It is unclear. Page 10, Lines 56-58 "...influence of several conditions of the effects of the interventions" - This section of the sentence is a bit wordy and hard to follow. Please clarify. Page 11, Line 28 - remove comma after "both" Page 11, Line 52 - change "every" to "all"

	Page 17, Line 56-58 - reword "such as setting and recipient, delivery as well as type of the intervention." This statement is hard to follow. Please reword for clarity. Page 19, Line 43 - place "i.e. benefits" in parentheses.

REVIEWER	Hung, Lillian Vancouver General Hospital
REVIEW RETURNED	07-Aug-2021

GENERAL COMMENTS	Thank you for the opportunity to review this protocol. Tech-based counseling in dementia is a timely and important topic. I have a few suggestions/ comments for considerations. 1. Introduction: It is important to articulate behavioral symptoms are indicating unmet needs so they are not simply consequences of dementia. Tech-based counselling can provide timely support to identify unmet needs and practical strategies to meet the person's needs. See https://journals.healio.com/doi/abs/10.3928/02793695-20160616-05 2. Please provide a justification for inclusion criteria. If you include informal caregiver at home, why not formal staff in LTC? Both formal and informal caregivers are partners in care. I am not clear why counselling as case management is excluded. P.8 Line 35 3. PPI and dissemination plan are well planned

REVIEWER	Sriram, Vimal Tokyo Institute of Technology
REVIEW RETURNED	14-Aug-2021

GENERAL COMMENTS	The purpose of this article for sharpening the focus on technological support for counseling to people with dementia and their informal caregivers is a very important and interesting research topic. This article protocol is very well-designed and is written with a clear description of procedures. Therefore, this reviewer has convinced that these authors will bring the significant result on the basis of this protocol.

REVIEWER	Sugihara, Taro Tokyo Institute of Technology
REVIEW RETURNED	14-Aug-2021

GENERAL COMMENTS	The purpose of this article for sharpening the focus on technological support for counseling to people with dementia and their informal caregivers is a very important and interesting research topic. This article protocol is very well-designed and is written with a clear description of procedures. Therefore, this reviewer has convinced that these authors will bring the significant result on the basis of this protocol.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1 comments **Dr. Kristopher Struckmeyer**	**Authors' comments**	**Page**
Page 9, Line 3 "Gender differences have..." - I would	We have deleted this sentence.	5

remove this sentence or go more in-depth regarding the known gender differences. In its current state, it seems out of place and does not seem to provide much information.		
Page 9, Line 9 "The worldwide COVID-19..." - change "e.g." to "related to". I noticed throughout the paper that you use "e.g." frequently. I would suggest placing those statements in parentheses "(e.g., lack of information...) or changing the wording to "related to" or something to similar to help with sentence flow.	We changed “e.g.” to “related to” and placed other examples in brackets.	5
Page 9, Line 33 "...to cope with emotional issue" - insert "an" between "with" and "emotional issue"	We have revised the sentence.	5
Page 10, Line 7 "...telephone counseling includes..." - This sentence seems out of place. I would either remove or move to another part of the paper.	We have removed the sentence.	6
Page 10, Line 18 "...for a more timely and individual..." - change individual to individualized	Changed as recommended.	6
Page 10, Line 35-39 "The authors concluded..." - delete ", which" and change influence to influenced.	Changed as recommended.	6
Page 10, Line 43 "completing strategies for services" - please clarify this statement. It is unclear.	We revised that sentence: “Due to the expanding number of related publications, one could assume an increasing demand to better support informal carers of people with dementia.”	6
Page 10, Lines 56-58 "...influence of several conditions of the effects of the interventions" - This section of the sentence is a bit wordy and hard to follow. Please clarify.	We have deleted this sentence.	6f.

Page 11, Line 28 - remove comma after "both"	Changed as recommended.	5,7
Page 11, Line 52 - change "every" to "all"	Changed as recommended.	8
Page 17, Line 56-58 - reword "such as setting and recipient, delivery as well as type of the intervention." This statement is hard to follow. Please reword for clarity.	We have revised this sentence and have placed the examples in brackets: «To assess the influence of different conditions on the effects of technology-based counselling (e.g., setting, recipient, facilitator, and type of intervention)…"	15
Page 19, Line 43 - place "i.e. benefits" in parentheses.	This paragraph was deleted based on the editor's recommendation.	16
Reviewer 2 comments **Dr. Lillian Hung**		
1. Introduction: It is important to articulate behavioral symptoms are indicating unmet needs so they are not simply consequences of dementia. Tech-based counselling can provide timely support to identify unmet needs and practical strategies to meet the person's needs. See https://journals.healio.com/doi/abs/10.3928/02793695-20160616-05	We have revised this section and added the proposed reference: "Common consequences of dementia are unmet needs, which can cause behavioural and psychological symptoms."	4
2. Please provide a justification for inclusion criteria. If you include informal caregiver at home, why not formal staff in LTC? Both formal and informal caregivers are partners in care. I am not clear why counselling as case management is excluded. P.8 Line 35	We have revised our inclusion criteria to be more precise. We now focus on informal caregivers that receive counselling by professionals without a restriction of setting. We	8ff.

	will exclude multi-component interventions that do not contain counselling as a main components or interventions which focus on care coordination (e.g., case management).	
Reviewer 3 comments **Mr. Vimal Sriram**		
2. One of the main issues that require clarification in this manuscript is – who is the population of interest. The authors identify in the methods section that they will include studies focussing on persons with dementia and their informal carers (Page 12 of 27 Lines 3-8) and in other sections such as conclusion (Page 19 of 27) they refer to including persons with dementia and informal carers. However, the description of intervention (Page 9 of 27), how the intervention might work (Pages 9,10 of 27) and primary outcome (Page 13 of 27) all describe the intervention only from the perspective of informal carers. The opening line of Objectives (Page 11 of 27) carries the quote “Overall, there is some evidence that informal carers would benefit from technology-based counselling” and the conclusion section (Page 19 of 27) starts with “Although there is some evidence that informal carers would benefit from technology-based counselling”. These discrepancies will confuse the reader about the population included in the	We have revised the background and methods to point out throughout the manuscript that we will include interventions for both peoples with dementia and their informal carers.	4ff.

systematic review – i.e. is it only informal carers or both persons with dementia and informal carers - and needs to be addressed and corrections made throughout the manuscript.		
3. Strengths and limitations: It will be useful to clarify which of the identified points (2-5) is a strength and which is a limitation?	We have clarified the strengths and limitations.	3
4. Strengths and limitations: Point 4 states stakeholder participation throughout the review process – however the patient and public involvement section and the methods section states representatives will be involved in the qualitative steps of the QCA (i.e. not in the search, data extraction and quality assessment of included studies), the authors should clarify this or rephrase the strength as stated in point 4.	We have clarified the stakeholder participation.	3
4. In the description of interventions on page 12 of 27, the authors state they will include talking therapy but exclude cognitive behaviour therapy (which is a form of talking therapy), it will be good to clarify why this is so.	We have revised our inclusion criteria and no longer use “talking therapy” to avoid confusion. We exclude standardised therapy procedures such as cognitive behaviour therapies.	8f.
5. The authors use the phrase “suffer from dementia” on page 8 of 27. It would be best to avoid such negative phrases about the experience of dementia.	The sentence has been revised.	4
6. There are a few typos in the manuscript that need to be fixed. For E.g., Page 8 of 27, Line 33 – should read as “Around half of all people with dementia need [assistance] with personal care”. Page 9 of 27, line 54 should be	Changed as recommended.	4,5,6,7

“mixed-methods study [uncovered]”. Page 10 of 27, line 43 should be “one could assume an increasing demand for [competing] strategies”. Page 11 of 27, Line 8 “[In addition] to telephone”.		
7. The authors need to provide references or links to software they describe in the manuscript such as TOSMANA and QCA add-in for Excel.	We have deleted the examples of the tools since we are not yet sure on that issue. In the final publication, we will report the tool with a reference.	14

8. I am not entirely sure what figure 1 as presented by the authors adds to the manuscript. It would be really useful for the reader (who may not be familiar with qualitative comparative analysis) to be shown the steps involved in detail.	We have deleted figure 1 and added the steps as headings to the corresponding paragraph on QCA.	12ff.
Editor(s)' Comments **Natasha Leeson**		
Conclusion sections are not part of journal formatting requirements for protocol articles. Please either remove the section or move the content of the section to another section within the manuscript.	We have added that part to the objectives.	16
In the abstract and methods, please include the month, as well as year, for the planned dates of coverage for each database.	We have added the month of search for all databases.	2,10
Please include, as a supplementary file, the precise, full search strategy (or strategies) for all databases, registers and websites, including any filters and limits used.	We have added the full search strategies for all databases as a supplementary file. “No use of filters and limiters” was added to the methods section.	Suppl., 10

VERSION 2 – REVIEW

REVIEWER	Struckmeyer, Kristopher Oklahoma State University, Human Development and Family Science
REVIEW RETURNED	19-Oct-2021

GENERAL COMMENTS	The edits that you made were great! This is a great paper!

REVIEWER	Hung, Lillian Vancouver General Hospital
REVIEW RETURNED	01-Nov-2021

GENERAL COMMENTS	This is a timely and important review study. Thank you for doing this work. I have a few suggestions for your consideration in revision. 1. Please provide explicit study questions 2. Consider applying a theoretical model/ framework (e.g., CFIR) to guide study planning and analysis. See https://cfirguide.org/ 3. More should be written about the relative advantages of the online platform for the specialized population. For example, virtual appointment avoids the stress and hurdles of in person visit. Families may struggle to leave the person with dementia at home to attend counselling sessions.The person with dementia may be more relaxed at a familiar home environment to receive counselling services. 4. It would be important to identify barriers and challenges that service users and providers face 5, If you only invite one person with dementia and one informal carer, is the 2-person group really going to function as an advisory board? Please describe their role in the study.

REVIEWER	Sriram, Vimal Tokyo Institute of Technology
REVIEW RETURNED	09-Oct-2021

GENERAL COMMENTS	I thank the authors for addressing all my comments for an earlier review of this manuscript. I would recommend the authors to check for spelling and gramatical errors - I have spotted a couple - e.g., abstract "CINAHL...Web of Science 'by' [until] April 2021. I do not have any other recommendations and will look forward to reading the results from this review.

VERSION 2 – AUTHOR RESPONSE

Reviewer 2 comments **Dr. Lillian Hung**	**Authors' comments**	**Page**
1. Please provide explicit study questions	We have added the review questions.	7
2. Consider applying a theoretical model/ framework (e.g., CFIR) to guide study planning and analysis. See https://cfirguide.org/	We have added a statement that we will develop a logic model to structure the conditions within the process of qualitative comparative analysis.	12
3. More should be written about the relative advantages of the online platform for the specialized population. For example, virtual appointment avoids the stress and hurdles of in person visit. Families may struggle to leave the person with dementia at home to attend counselling sessions. The person with dementia may be more relaxed at a familiar home environment to receive counselling services.	We have revised the background accordingly.	5-6
4. It would be important to identify barriers and challenges that service users and providers face	We have clarified these issues in our objectives and "Data synthesis" section.	7,12

5, If you only invite one person with dementia and one informal carer, is the 2-person group really going to function as an advisory board? Please describe their role in the study.	The advisory board will consist of two or three people. We have clarified their role in the section “Patient and Public Involvement”.	15
Reviewer 3 comments **Mr. Vimal Sriram**		
I thank the authors for addressing all my comments for an earlier review of this manuscript. I would recommend the authors to check for spelling and gramatical errors - I have spotted a couple - e.g., abstract "CINAHL...Web of Science 'by' [until] April 2021. I do not have any other recommendations and will look forward to reading the results from this review.	We have made language revisions throughout the manuscript.	-